PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME										
CT GROUP	1	6	S			INDEX NUMBER				

BIOLOGY 9744/02

Paper 2 Structured Questions

15 September 2017
2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.

All workings and appropriate units must be shown.

The number of marks is given in brackets [ ] at the start of each question or part question.

For Examiner's Use	
1	**/ 8**
2	**/ 11**
3	**/ 18**
4	**/ 8**
5	**/ 12**
6	**/ 8**
7	**/ 12**
8	**/ 11**
9	**/ 12**
Total	**/ 100**

Answer **all** questions in this paper.

Question 1 [8 marks]

Fig. 1.1 illustrates an electron micrograph of a plant cell.

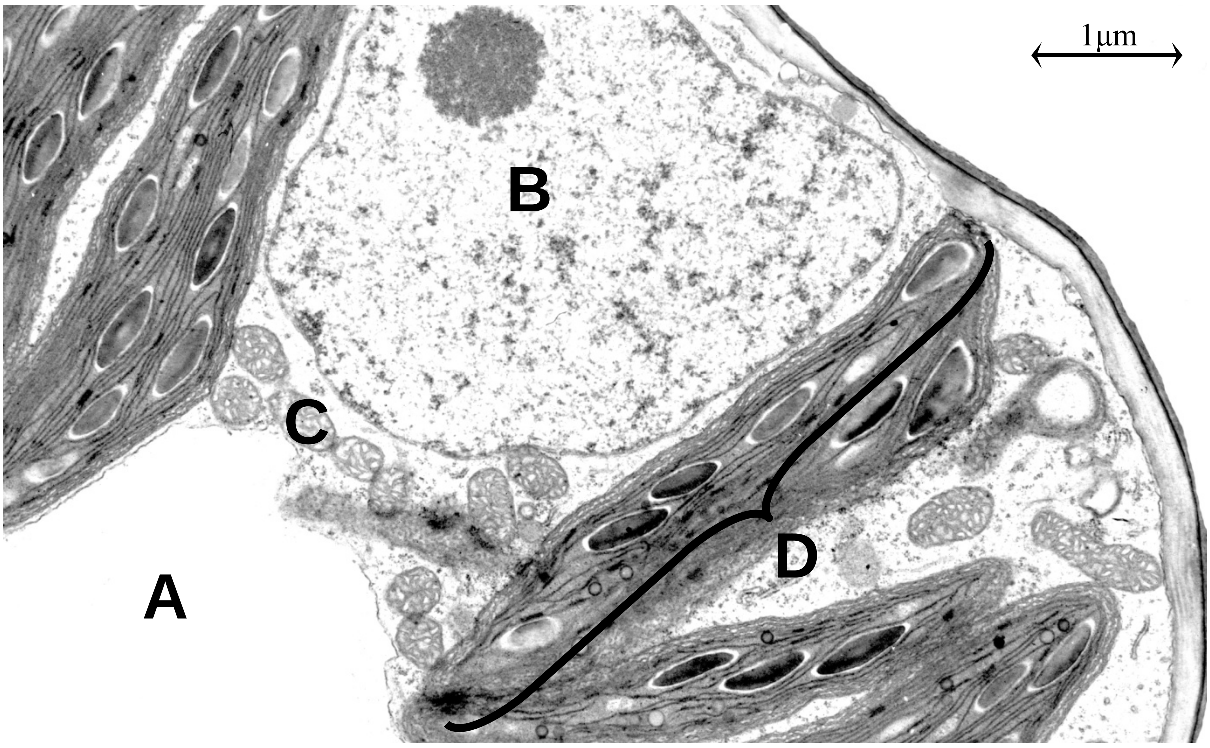

Fig. 1.1

(a) With reference to **Fig. 1.1**, identify organelles labelled **A** and **B** and justify your answers. [4]

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(b) Explain how the structural features of organelles **C** and **D** can contribute to its role in the plant cell. [2]

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Fig. 1.2 is a diagram of the Golgi body, an organelle found in most eukaryotic cells.

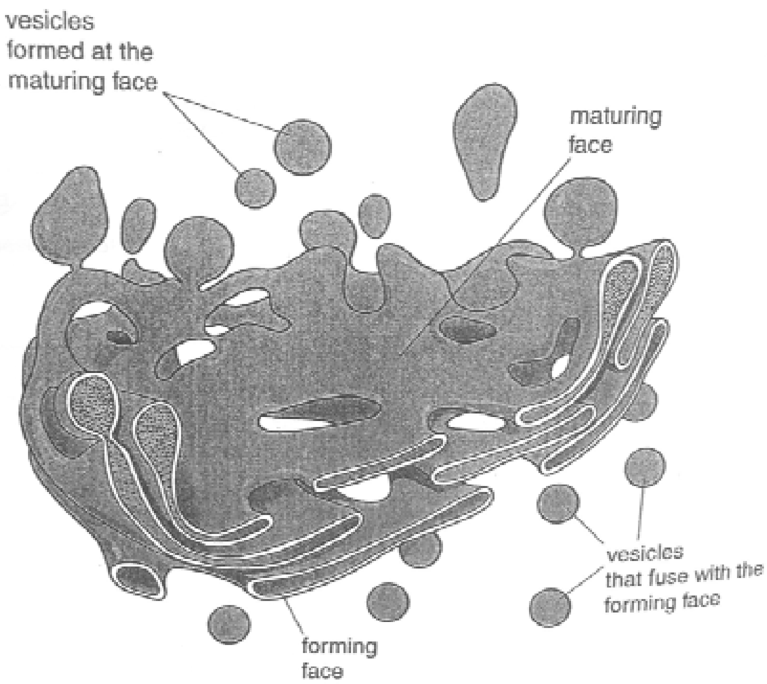

Fig. 1.2

(c) Besides serving as secretory vesicles, outline another role for the vesicles that are formed at the maturing face of the Golgi body. [2]

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Question 2 [11 marks]

Fig. 2.1 shows the structure of protease.

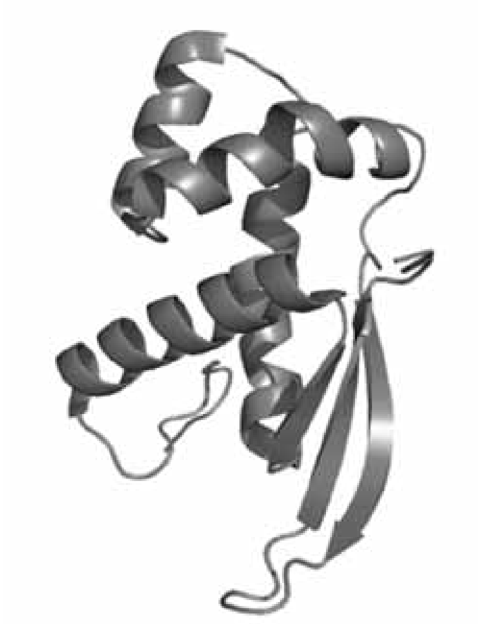

Fig. 2.1

(a) With reference to **Fig. 2.1**,

(i) explain why an enzyme acts only on a specific substrate. [2]

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(ii) explain what determines the three-dimensional conformation of protease. [2]

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Sorghum is a staple food in Africa, but the major storage protein that it contains, kafirin, is not easily digested by protease enzymes. Upon heating, kafirin can undergo unfolding as shown in **Fig. 2.2**.

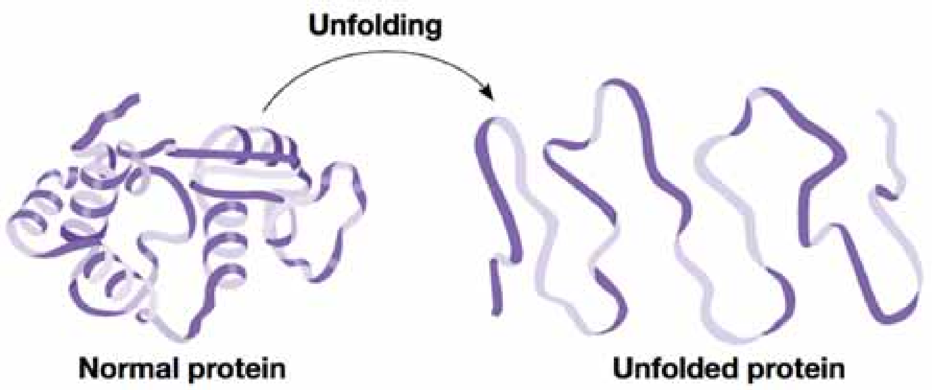

Fig. 2.2

The digestibility of the protein in two varieties of sorghum was measured when raw, after cooking with and without acid. Digestibility was measured as the percentage of the protein that would be broken down to amino acids during digestion.

The results are shown in **Fig. 2.3**.

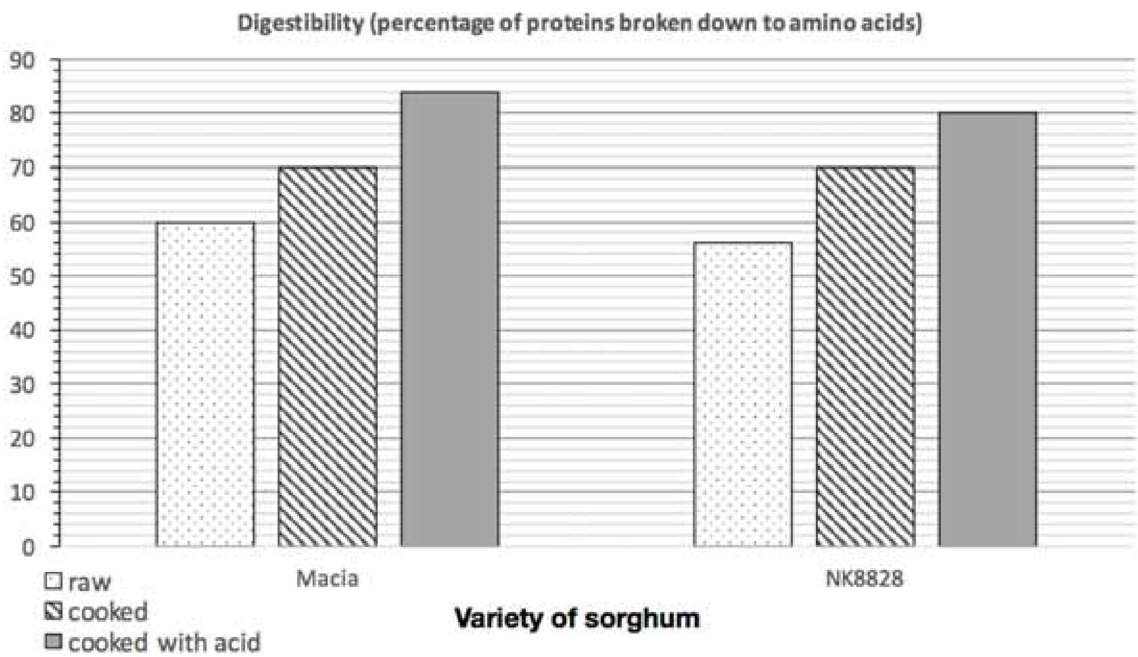

Fig. 2.3

(b) With reference to **Fig. 2.3**,

(i) <u>**difference 1**</u>: compare the digestibility of raw and cooked sorghum protein. [1]

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(ii) **difference 2**: compare the digestibility of cooked sorghum with and without acid. [1]

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(iii) and using **Fig. 2.2** with your knowledge of protein structure and enzyme activity, account for the two differences described in parts **(i)** and **(ii)**. [3]

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(c) Describe how the bond holding the two amino acids together may be broken to release the two amino acids. [2]

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Question 3 [18 marks]

The production of membrane-bound or secreted immunoglobulin M (IgM) depends on whether the B cell is activated. Naïve B cells produce membrane-bound IgM while activated B cells produce soluble IgM that is secreted out of the cell.

In naïve B cells, the gene coding for heavy chain of IgM is expressed to produce a long RNA transcript, while the same gene codes for a shorter RNA transcript when the cell is activated (see **Fig. 3.1**). The two different types of RNA are processed differently, resulting in two different types of antibodies upon translation.

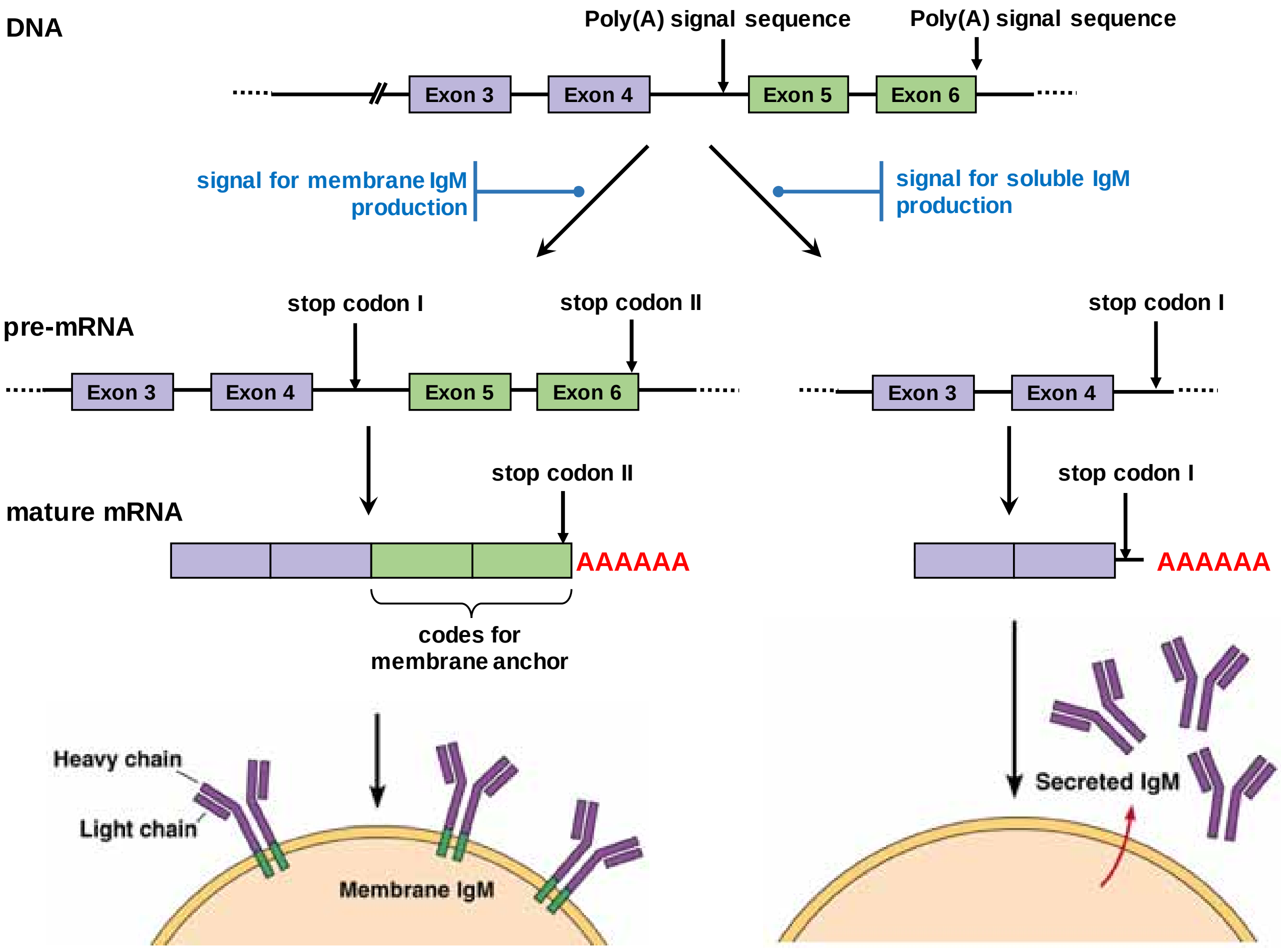

Fig. 3.1

Fig. 3.2 shows the region of DNA between exons 4 and 5.

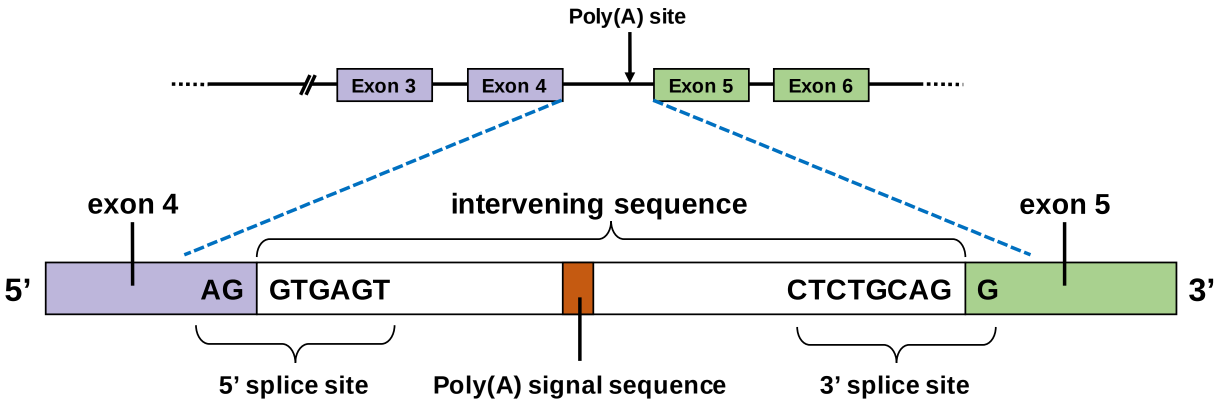

Fig. 3.2

(a) With reference to **Fig. 3.1**,

(i) describe how the different types of mRNA produced determine how IgM is membrane-bound or secreted out of the cell. [2]

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(ii) describe how IgM is held in the membrane of B cells. [2]

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(iii) The membrane anchor in membrane IgM is reported to be associated with several proteins on the cytoplasmic side.

Suggest the function of these associated proteins. [1]

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(b) The activation of a naïve B cell involved the recognition of specific antigens, and mediated by helper T cells.

With reference to **Fig. 3.1**,

(i) describe the events occurring at the promoter of the IgM heavy chain gene locus in both naïve and activated B cells. [2]

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(ii) describe how transcription is terminated when the B cell is activated. [2]

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(iii) explain how the mature mRNA produced is relatively more stable than the pre-mRNA. [3]

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(c) With reference to both **Fig. 3.1** and **3.2**, explain the importance of the 5' and 3' splice site. [2]

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In order to understand the process of translation, researchers isolated a clone of activated B cells and introduced small interfering RNA (siRNA) into these cells. siRNA is an RNA molecule that can be designed and produced *in vitro*.

In this study, the siRNA introduced was complementary to a short segment of RNA on the 5' end of the mature mRNA. It was found that these activated B cells were unable to produce soluble IgM antibodies.

(d) **(i)** Describe the structural features of the monomers in siRNA. [2]

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(ii) Suggest why the introduction of the siRNA did not result in production of IgM antibodies. [2]

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Question 4 [8 marks]

Colon cancer begins when a number of epithelial cells lining the colon proliferate excessively due to changes in the several genes. **Fig. 4.1** shows the progression of colon cancer from a normal epithelium to a metastatic colon cancer.

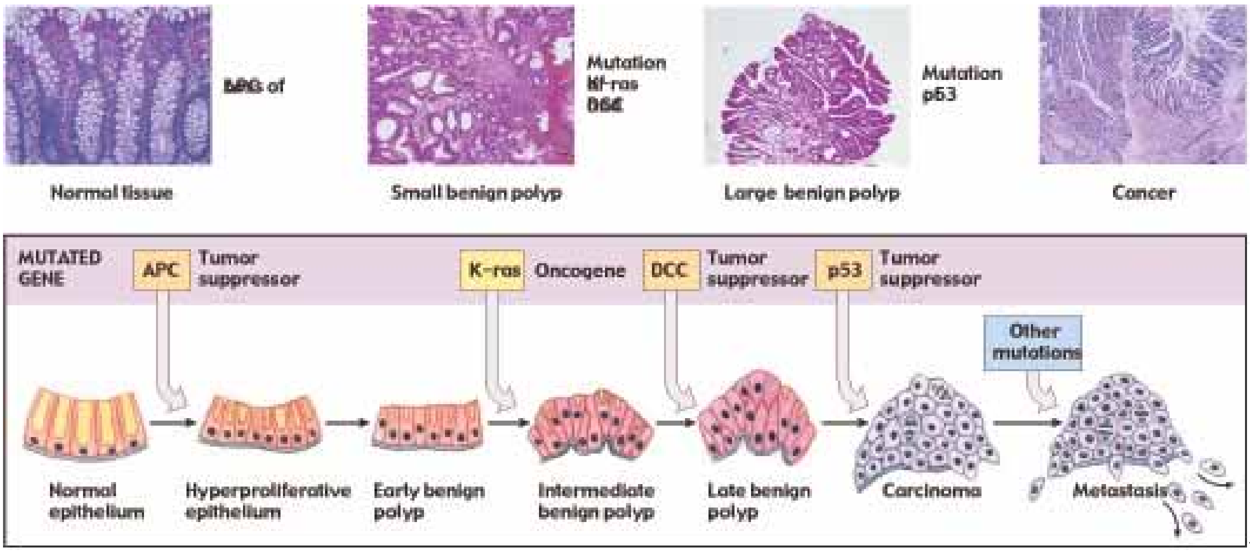

Fig. 4.1

(a) With reference to **Fig. 4.1**, explain how this supports the multi-step model of cancer. [2]

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The effectiveness of anti-cancer drugs may be determined by growing different tumours in culture.

The effectiveness of two drugs on two human tumours (**A** and **B**) from different tissues was assessed. The two drugs, T138067 and vinblastine, were added to the tumours in the culture on days 5, 12, and 19. The volumes of the tumours were compared with the volumes of tumours that were not treated with any drugs.

The results are shown in **Fig. 4.2**.

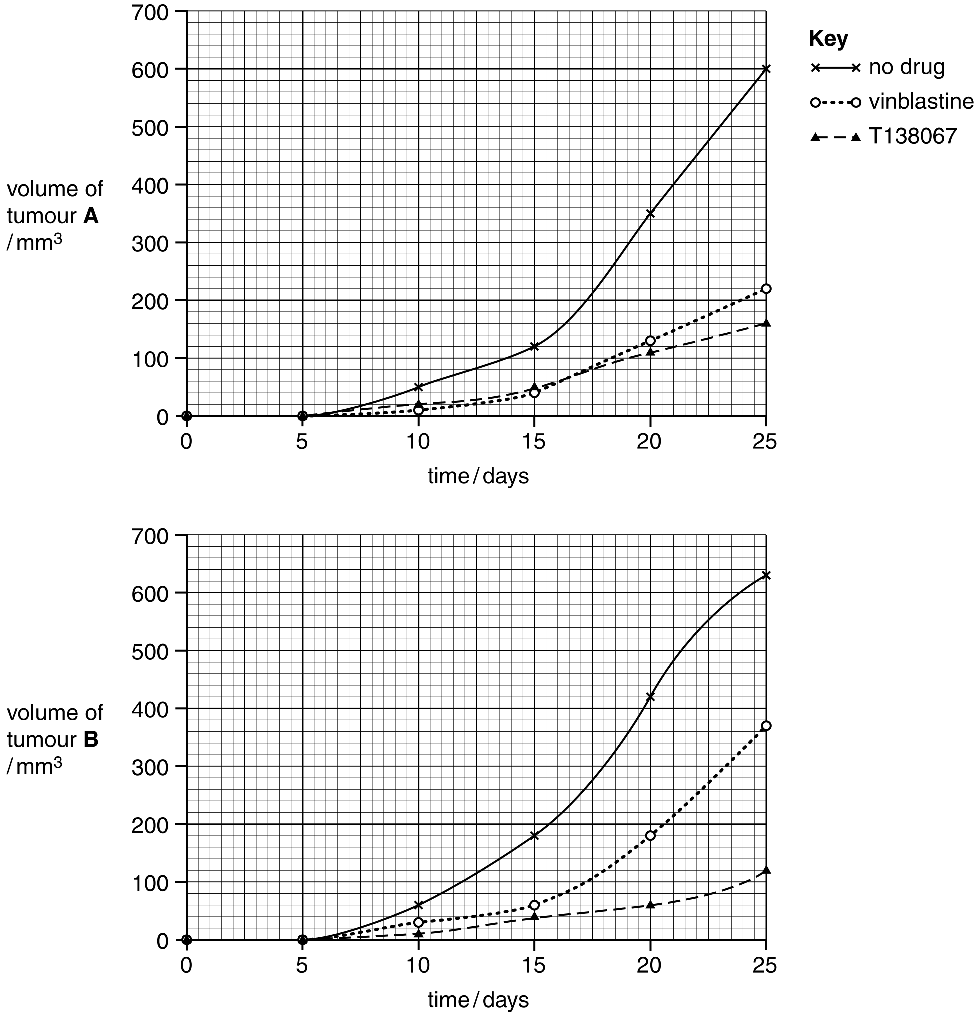

Fig. 4.2

(b) Using the data in **Fig. 4.2,** compare the effectiveness of the two drugs used to treat the tumours. [3]

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(c) Vinblastine disrupts the formation of the spindle apparatus during mitosis.

Explain how vinblastine has its effect as an anti-cancer drug. [3]

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Question 5 [12 marks]

Fig. 5.1 shows an electron micrograph of H1N1.

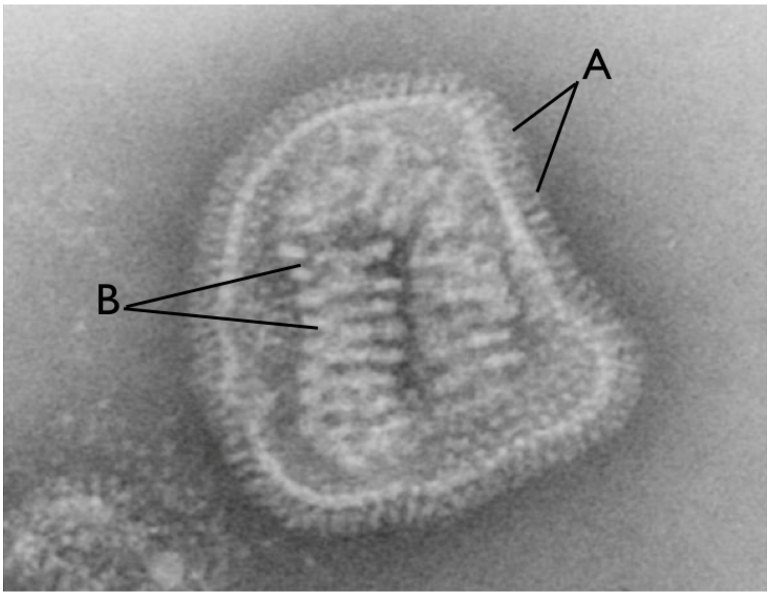

Fig. 5.1

(a) Identify the components labelled **A** and **B**. [3]

A: …………………………………………………………………

B: …………………………………………………………………

(b) RNA-dependent RNA polymerase can be found in this virion while reverse transcriptase can be found in retrovirus.

Besides the type of virus, describe one similarity and two differences between RNA-dependent RNA polymerase and reverse transcriptase. [3]

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Oseltamivir, better known under its trademark name as Tamiflu, is a recommended medication for H1N1. Its mechanism of action is illustrated in **Fig. 5.2** and **Fig. 5.3**.

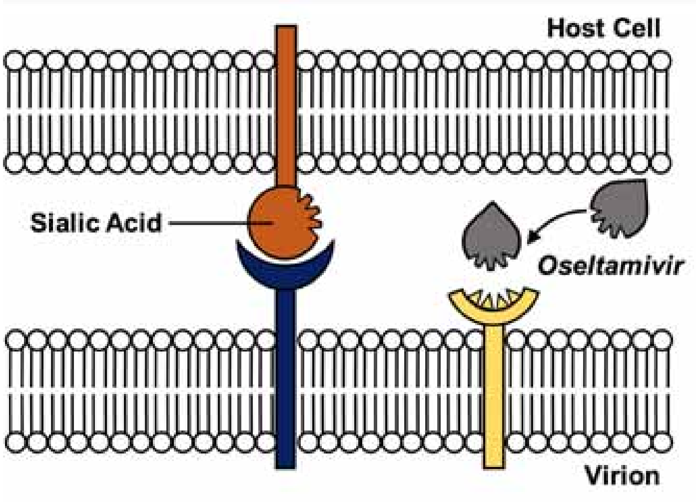

Fig. 5.2

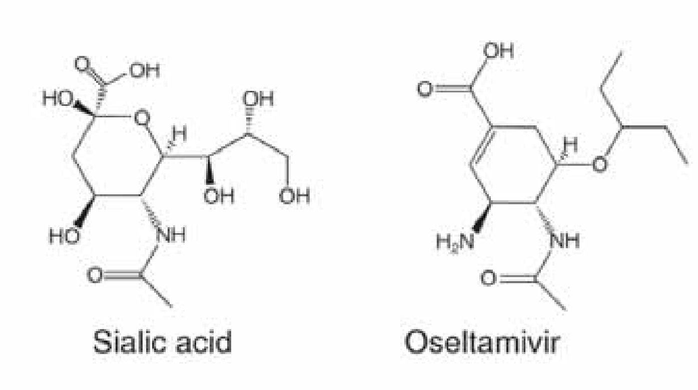

Fig. 5.3

(c) Explain how oseltamivir affects the reproduction cycle of H1N1. [2]

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Many studies were conducted to determine the effectiveness of Oseltamivir.

In one of these studies, the effects of oseltamivir were studied on patients infected with influenza virus. These patients were randomly separated into two groups – one group administered with a placebo (a pseudo-drug with no effect) while the other group was administered oseltamivir. Many symptoms were evaluated, with cough being one of the symptoms represented in **Table 5.4**.

Table 5.4

Symptoms	**Study Groups**	
	Placebo (n=129)	**Oseltamivir, 75mg (n=124)**
Cough		
Duration, hr	55	31
Severity, arbitrary units	110	67

Facing the wave of flu cases, oseltamivir was stocked up in many medical facilities across countries where there was widespread administration of this drug. Thereafter, there were reports of side effects that included hallucination, and a flurry of questions rose to challenge the validity of studies such as the one above.

(d) Suggest what could have been done in the above study to increase the confidence of the effects of oseltamivir. [1]

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(e) In a separate case, a patient was treated for both H1N1 and H3N3. During this treatment, he was infected with H3N3. Upon testing, it was found that his blood now contains H3N3 and a new strain of virus, H1N3.

Explain how this new strain could have arisen. [3]

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Question 6 [8 marks]

Fig. 6.1 shows two bacterial cells, bacterium **A** and bacterium **B** involved in process **X**.

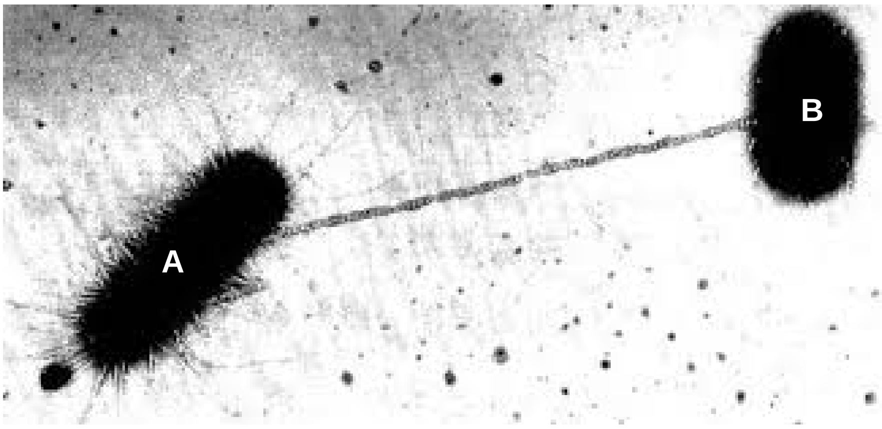

Fig. 6.1

(a) Describe the process **X** and elaborate on its significance. [3]

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Fig. 6.2 shows a section of an electron micrograph of bacterium **A** in **Fig. 6.1**.

Fig. 6.2

(b) Explain the role of structure **C**. [3]

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(c) State two ways in which structure **C** differs from the bacterial chromosome. [2]

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Question 7 [12 marks]

An experiment was conducted to investigate how various factors affect the rate of photosynthesis in cabbage. **Fig. 7.1** below shows the results of the experiments conducted.

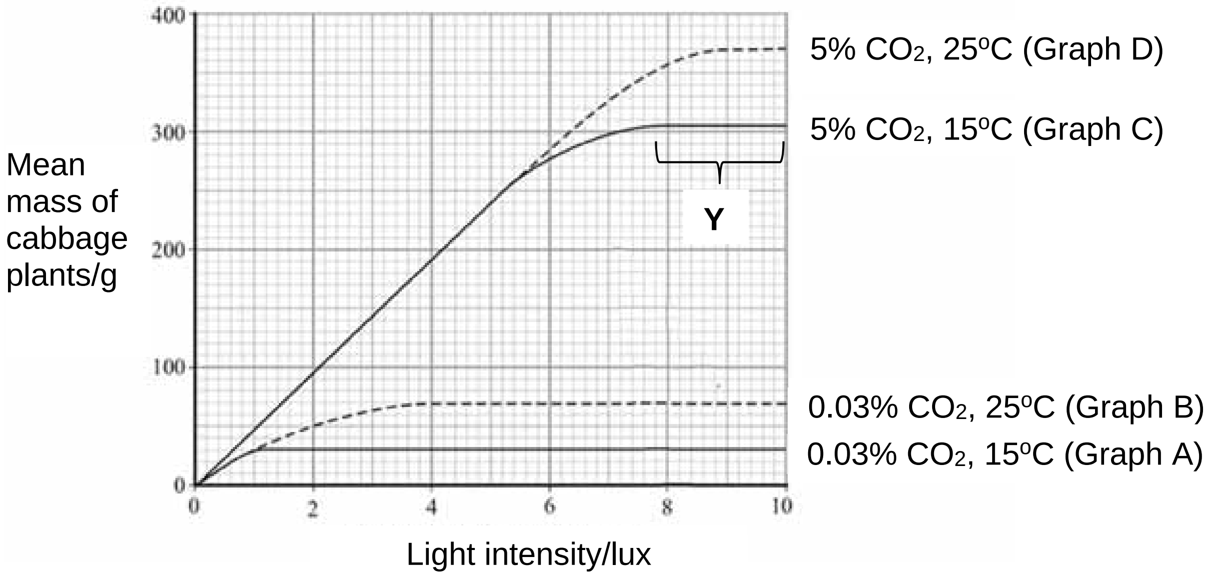

Fig. 7.1

(a) With reference to **Fig. 7.1**,

(i) state the best conditions for the growth of cabbage. [1]

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(ii) explain the region marked **Y**. [2]

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(iii) describe and explain the effect of increasing carbon dioxide concentration on the mean mass of cabbage at 25 ºC. [3]

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(iv) The average carbon dioxide content of the natural environment is 0.035%. Using this fact, and the information given in **Fig. 7.1**, what conclusion can be made about how carbon dioxide affects rate of photosynthesis in the natural environment? [2]

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(b) While photosynthesis is the process by which carbon dioxide and water are used as starting materials for the synthesis of glucose using light energy, respiration involves releasing chemical energy in organic molecules such as glucose by oxidation and made available to living cells in the form of ATP. In particular, the yield of ATP under aerobic and anaerobic respiration are very different.

Explain the small yield of ATP from anaerobic respiration in both yeast and animals. [4]

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Question 8 [11 marks]

In cattle, coats may be solid white, solid brown, or beige. When true breeding solid whites are mated with true-breeding solid brown, the F1 generation consists of all solid white individuals. Mating among the F1 generation have resulted in the following ratio:

23 solid white
6 beige
3 solid brown

(a) **(i)** Draw a genetic diagram of the above described cross. [4]

(ii) Explain what is meant by 'epistasis' in this context. [3]

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(iii) The chi-squared test was later performed on the F2 data.

Explain what a chi-squared test is used for. [1]

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Marfan syndrome is a genetic disorder of the connective tissues. People with Marfan tend to be tall, and thin, with long arms, legs, fingers and toes. The most serious complications involve the heart and aorta.

Fig. 8.1 shows the inheritance of Marfan syndrome over three generations in an extended family.

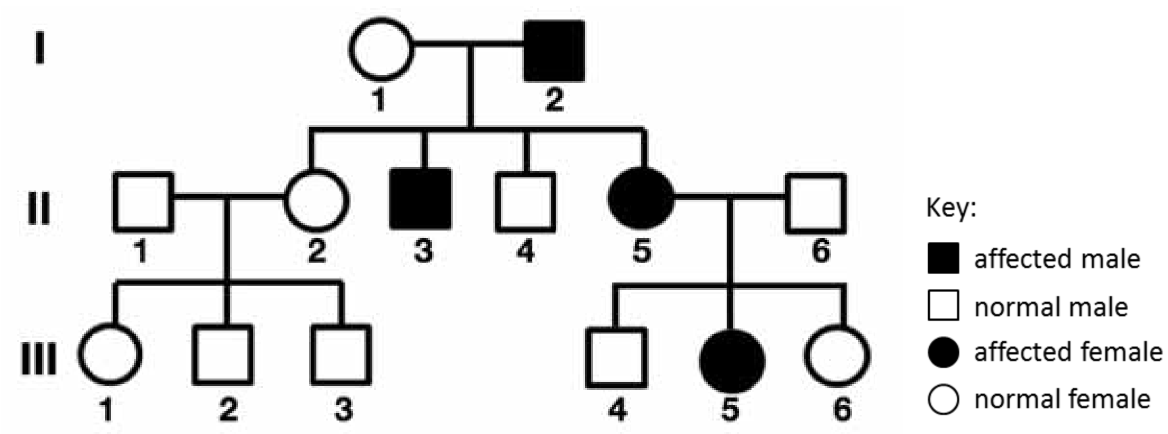

Fig. 8.1

(b) Explain the mode of inheritance of this disease that best explains the pedigree shown above. [3]

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Question 9 [12 marks]

The sensitivity of bacteria to antibiotics can be tested using the disc diffusion method. A small number of bacteria is spread onto agar culture plates and then filter discs impregnated with antibiotics are pressed onto the surface of the agar. The plates are incubated. Bacteria grow as a 'lawn' across the agar, but a circular zone (the zone of inhibition) appears around any disc where bacterial growth is inhibited.

Two species of bacteria, **A** and **B**, were grown on separate culture plates in the presence of three types of filter paper disc:

1 – no antibiotics (control)
2 – penicillin V, a natural penicillin
3 – carboxypenicillin, a synthetic penicillin.

The appearance of the incubated plates is shown in **Fig. 9.1**.

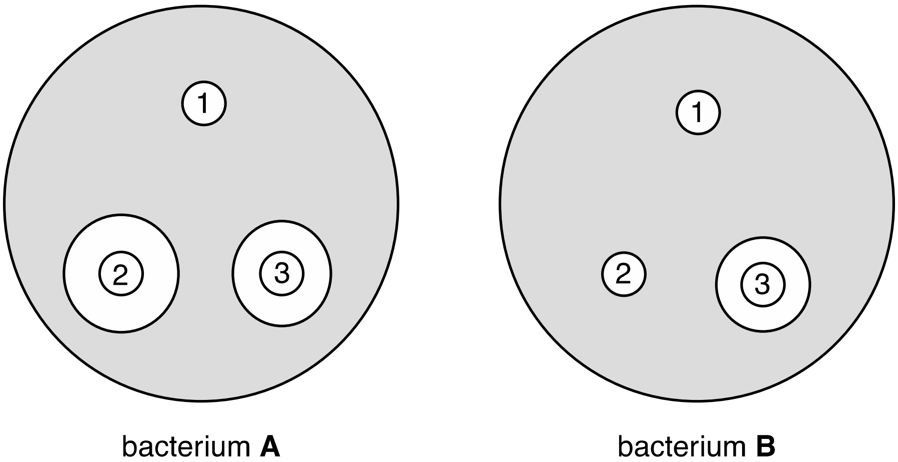

Fig. 9.1

(a) With reference to **Fig. 9.1**, explain the effects of penicillin V on bacterium **A**. [3]

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Bacteria **A** and **B** have different outer layers, as shown in **Fig. 9.2**.

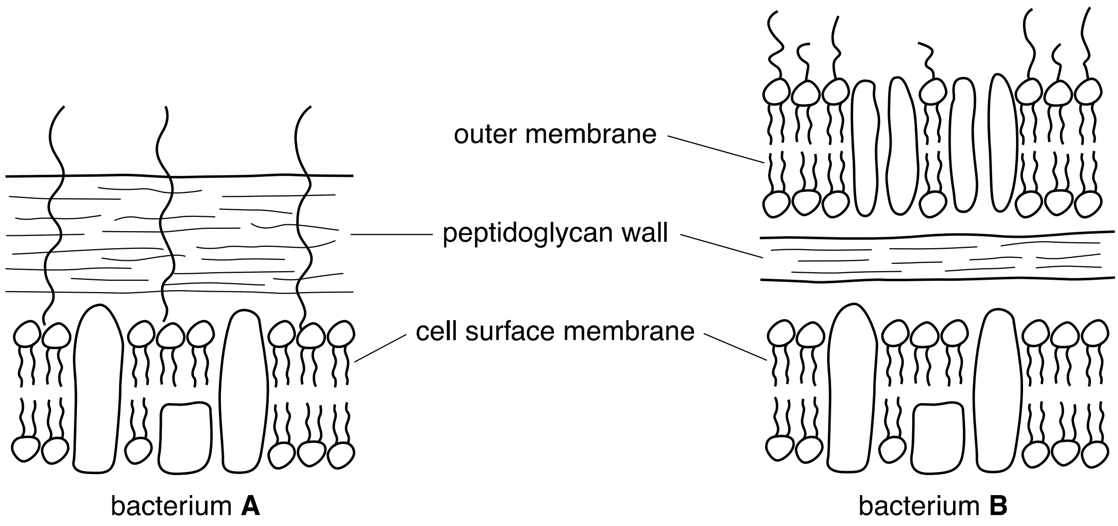

Fig. 9.2

(b) With reference to **Fig. 9.1** and **9.2**,

(i) describe how the outer layers of bacterium **B** differ from those of bacterium **A**. [2]

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(ii) account the different effects of penicillin V on bacteria **A** and **B**. [3]

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(iii) suggest how the synthetic penicillin, carboxypenicillin, is able to affect the growth of bacterium **B**. [2]

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(c) Outline two other ways in which antibiotics are effective against diseases caused by bacteria. [2]

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